Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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9 Disproof

Each of the following statements is either true or false. If a statement is true, prove it. If a statement is false, disprove it. These exercises are cumulative, covering all topics addressed in Chapters 1-9.

2. For every natural number n, the integer $2n^2 - 4n + 31$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 31$. Then $2n^2 - 4n + 31 = 2(31^2) - 4(31) + 31 = 1829 = 31 \cdot 59$, which is not prime. □

4. For every natural number n, the integer $n^2 + 17n + 17$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 1$. Then $n^2 + 17n + 17 = 1^2 + 17(1) + 17 = 35 = 7 \cdot 5$, which is not prime. □

6. If A, B, C and D are sets, then $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Proof. Observe the following sequence of equalities.

$$
\begin{aligned}
&(A \times B) \cap (C \times D) \\
&\quad = \{(x, y) : (x, y) \in A \times B \land (x, y) \in C \times D\} && \text{(definition of } \cap\text{)} \\
&\quad = \{(x, y) : (x \in A \land y \in B) \land (x \in C \land y \in D)\} && \text{(definition of } \times\text{)} \\
&\quad = \{(x, y) : (x \in A \land x \in C) \land (y \in B \land y \in D)\} && \text{(regrouping)} \\
&\quad = \{(x, y) : x \in A \cap C \land y \in B \cap D\} && \text{(definition of } \cap\text{)} \\
&\quad = \{(x, y) : (x, y) \in (A \cap C) \times (B \cap D)\} && \text{(definition of } \times\text{)} \\
&\quad = (A \cap C) \times (B \cap D) && \text{(definition of set-builder)}
\end{aligned}
$$

□

8. If A, B and C are sets, then $A - (B \cup C) = (A - B) \cup (A - C)$.

Disproof. The statement is false. For a counterexample, let $A = \{1, 2\}$, $B = \{1\}$ and $C = \{2\}$. Then $A - (B \cup C) = \{1, 2\} - \{1, 2\} = \emptyset$, but $(A - B) \cup (A - C) = \{2\} \cup \{1\} = \{1, 2\}$. Since $\emptyset \neq \{1, 2\}$, the two sides are not equal. $\square$

10. If A and B are sets and $A \cap B = \emptyset$, then $\mathcal{P}(A) - \mathcal{P}(B) \subseteq \mathcal{P}(A - B)$.

Proof. Suppose $A \cap B = \emptyset$. Let $X \in \mathcal{P}(A) - \mathcal{P}(B)$. Then $X \in \mathcal{P}(A)$ and $X \notin \mathcal{P}(B)$, so $X \subseteq A$. Now we claim $A - B = A$. Let $a \in A - B$. Then $a \in A$ and $a \notin B$. Thus every element of $A - B$ is an element of A, so $A - B \subseteq A$. Conversely, let $a \in A$. If $a \in B$, then $a \in A$ and $a \in B$, so $a \in A \cap B = \emptyset$, which is impossible. Thus $a \notin B$. So we have $a \in A$ and $a \notin B$, which means $a \in A - B$. Hence $A \subseteq A - B$. Therefore $A - B = A$. So we have $X \subseteq A = A - B$, thus $X \in \mathcal{P}(A - B)$. Since X was arbitrary, $\mathcal{P}(A) - \mathcal{P}(B) \subseteq \mathcal{P}(A - B)$. $\square$